# Soroush Mahdi

soroush-mim
soroush-mim.github.io
soroush-mahdi
soroushm1377@gmail.com
soroushm@aut.ac.ir
+98 935 914 3257

## Education

**Master of Science in Artificial Intelligence** — *Tehran, Iran*
*AmirKabir University of Technology (Tehran Polytechnic)* — *Sep 2020 – Sep 2023*
*Ranked **7th** among 50 M.Sc. students*
*Cumulative **GPA: 4/4***

**Bachelor of Computer Engineering** — *Hamedan, Iran*
*Bu-Ali Sina University* — *Sep 2016 – Sep 2020*
*Cumulative GPA: 3.17*

## Research Interests

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|---|---|
| **Computer Vision** | **Vision-Language Models (VLMs)** |
| **Generative AI** | **Medical Image Analysis** |

## Publications

1. **Mahdi, S.**, Ayar, F., Javanmardi, E., Tsukada, M., Javanmardi, M., "Evict3R: Training-Free Token Eviction for Memory-Bounded Streaming Visual Geometry Transformers". arXiv:2509.17650, 2025.
2. Dastmalchi, H., **Mahdi, S.**, Alimoradijazi, M., Tamjidi, M., Saberi, M., *et al.*, "MODE-TTA: Multi-Mode Density Modeling for Test-Time Adaptation of 3D Vision-Language Foundation Models." Submitted to ECCV, 2026.
3. **Mahdi, S.**, Nabati, S., Dehghanian, Z., Amirmazlaghani, M., "MemLoss: Enhancing Adversarial Training with Momentum-Based Memory". arXiv:2510.09105, 2025.

## Honors & Awards

- **1st** Place in The Iran Annual AI Award, **focusing on brain abnormality diagnosis using MRI DICOM images**, Cognitive Sciences and Technologies Council, Tehran, Iran, 2024.
- **2nd** Place in The National fNIRS Data Analysis Competition, National Brain Mapping Laboratory, Tehran, Iran, 2022.
- Ranked **61st** (AI) and **36th** (Algorithms) among 15,000 participants in the Iranian National M.Sc. Entrance Exam, 2020.
- Ranked **22nd** Place in National Collegiate Scientific Olympiad in Computer Engineering, Iran National Organization of Educational Testing, Tehran, Iran, 2020.
- Selected as a member of Bu-Ali Sina University's team for the West Asia Regional ACM-ICPC Contest, Iran, 2019 & 2017.

## Academic Experiences

**Research Assistant** — *Jan 2025 – Present*
*University of Sydney* — *Remote*
*Under the supervision of Ali Cheraghian*

- Co-developed **MODE-TTA**, a test-time adaptation framework for 3D vision–language foundation models, including class-conditional density modeling with Gaussian Mixture Models (GMMs).
- Implemented the full pipeline and conducted extensive experiments on LiDAR-based benchmarks.

**Research Assistant** — *June 2025 – Present*
*Autonomous & Intelligent Systems Lab, Amirkabir University of Technology* — *Tehran, Iran*
*Under the supervision of Mahdi Javanmardi*

- Working on 3D Computer Vision and Vision-Language Models.

- Developed Evict3r, a novel token eviction method for deep foundation SFM models, reducing memory usage by over 50% in large-scale 3D scene reconstruction without performance loss.

**Thesis Based M.Sc. Student and Research Assistant** *June 2021 – Present*
*Statistical Data Analysis Laboratory, Amirkabir University of Technology* *Tehran, Iran*
*Under the supervision of Maryam Amirmazlaghani*

- Thesis title: An approach for improving the robustness of deep neural network image classifiers against adversarial examples.
- Proposed **MemLoss**, a memory-based adversarial training method that improved both clean and robust accuracy.

## *Industry Experiences*

**Data Scientist** *July 2024 – June 2025*
*Iran Credit Scoring* *Tehran, Iran*

- Worked as a data scientist in the R&D team.
- Developed a machine learning–based credit scoring model for evaluating cheque creditworthiness.

## *Academic Projects*

- **Evict3r** - GitHub
  Designed a training-free token eviction mechanism reducing memory usage by over 50% in 3D scene reconstruction.
- **MEMLoss** - GitHub
  Proposed a memory-based adversarial training framework that reuses past adversarial examples to improve both clean and robust accuracy in image classification.
- **PSO-SGD Hybrid Optimizer** - GitHub
  Developed a custom PyTorch optimizer by designing an optimization algorithm that combines SGD and PSO.
- **multivariate HMM** - GitHub
  Implemented a range of algorithms for Hidden Markov Models (HHMs) in the multivariate case from scratch.

## *Teaching Experience*

- **Graduate Courses – Amirkabir University of Technology:** Statistical Machine Learning (Spring 2023); Machine Vision (Fall 2022)
- **Undergraduate Courses – Bu-Ali Sina University:** Algorithm Design (2020); Artificial Intelligence & Expert Systems (2019); Data Structures (2018)

## *Professional Skills*

- **Programming:** Python, C++, R
- **Frameworks:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas, scikit-learn, JAX
- **Tools:** Git, LaTeX, Visual Studio Code
- **Languages:** English (TOEFL iBT 103/120), Persian (Native)

## *Major Courses*

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|---|---|---|---|
| **Machine Vision** | 19.9/20 | **Stochastic Processes** | 19.28/20 |
| **Machine Learning** | 18.6/20 | **Neural Networks** | 16.8/20 |

## *References*

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|---|---|---|
| Mahdi Javanmardi (Assistant Professor) | mjavan@aut.ac.ir | Homepage |
| Ali Cheraghian (Adjunct Lecturer) | ali.cheraghian@mq.edu.au | Homepage |

More references are available upon request.